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<u>Título/Title</u> Program ETUR User's Manual Version 3				
<u>Palabras clave/Key words</u> MANUALS / BOLTED JOINTS / THERMAL ANALYSIS / COMPUTER PROGRAMAS				Clasificación acceso Access class P1 Registro de Revisiones Revision Record Pag. 2
<u>Resumen/Summary</u> The purpose of this technical report is to define the way to use the ETUR program. This program calculates the loads of a fastened single shear joint between materials with different thermal expansion coefficient, subjected to an increase of temperature. The stiffness of the joints is calculated according to the Carbon Fibre Riveted Joints Manual.				
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1.- Technical sheet of the program

Technical Note **ME-T-NT-110052**

PROGRAM

ETUR

REFERENCE

DATE

February 2011

OBJECTIVE

This program calculates the loads of a fastened single shear joint between materials with different thermal expansion coefficient, subjected to an increase of temperature.

LANGUAGE

Fortran 77

COMPUTER

Multivac

NUMBER OF LINES (APROX.)

1400

EXECUTION MODE

Interactive

AUTHOR

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VERSION

1.0

Darío Fernández Villalba

3.0

ADDITIONAL DOCUMENTS

Technical Note: NT-T-ADCE-025/92.
Theoretical Manual of the Program ETUR, Version 1.0

2.- Introduction

This program calculates the loads of a fastened single shear joint between materials with different thermal expansion subjected to an increase of temperature. The stiffness of the joints is calculated according to the Carbon Fibre Riveted Joints Manual.

The considered geometry of the joint is shown on next figure:

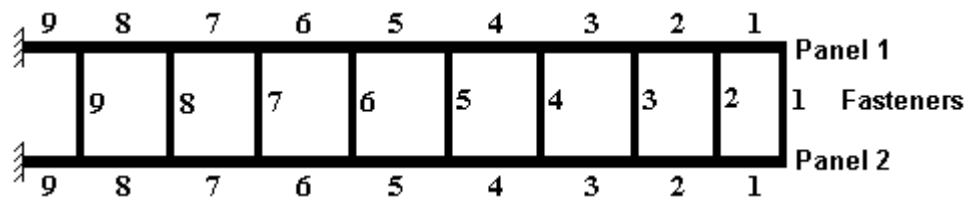


Figure 1: General configuration.

The program calculates for each fastener the following values:

- Stiffness and shear load of each fastener.
- Force on each one of the elements of the joint introduced by fasteners.
- Nodal displacements on the elements of the joint.

The program only works with single shear linear joints that have free displacement in one direction. It implies the assumption of different hypotheses according to the type of joint to analyse. The purpose of these hypotheses is to reduce, under an engineering viewpoint that is coherent with the type of analysis, the complex real problem to one or other problems that can be analysed with the current program. The following are some hypotheses commonly assumed that will be explained when used to present the examples analysed in this report.

- A typical problem of displacements on both directions is transformed in two problems of displacement on one direction only, dividing the original problem by the symmetry plane, if it exists, or by the section where the displacement can be considered null. In this last case, it is possible to later verify if the hypothesis made is accurate comparing the displacements obtained for both problems in which the original problem was divided.
- A double shear joint problem becomes two simple shear joint problems by assigning – from the original central member- to each one of the two simplified resulting joints, the corresponding thickness, depending on the stiffness associated to the new joints. In the simple case that the two external members have the same thickness and are made of the same material (the same elastic module), the thickness of the central member assigned to the two new joints will be half the original one.

Differences between versions:

- Version 1.0:

Initial version developed in Fortran 77 for Digital VAX computer.

- Version 2.0:

The program is adapted to the new Sun Solaris computer.

- Version 2.1:

The program is improved and the output file is translated to English.

- Version 3.0:

Now, the user can choose the language for screen dialogs and for the output file, between Spanish and English.

The data flow of the program is modified. Now, the input data as well as the output results are provided by means of files:

- Input file: With the input data grouped in the same way as for other applications.
- Output file: With all the input data and output results in printable form.

Since the format and arrangement of the input file is completely different from previous version, input files are not compatible between versions.

The program source is restructured to allow the creation of a Dynamic Link Library DLL for Windows operating system.

In parallel with this, a Windows application has been developed as Graphic User Interface GUI for the program ETUR. This application uses the same input file and provides the same output results. The analysis is performed by calling to the DLL, compiled from the same source as the official Fortran version.

3.- Execution of the program

Prior to the program execution, it is necessary to create the input data file, as described in section 4. The program can be then executed by typing its **full path name**.

If the software symbol table has been previously assigned (consult with the System Manager in order to know how to assign this symbol table); it can be executed by typing the command:

```
$etur
```

The program asks for the language to use, the name of the input file and the name of the two output files.

```
Idioma / Language:
=====

 1 ... ESPAÑOL

 2 ... ENGLISH

Idioma / Language: 2

Name of the input file:   Example.dat

Name of the output file: Example.txt
```

Details of the input and output files are provided in next sections. The CPU time for calculation is negligible.

It is possible to execute the program with a single command, including the input and output filenames and the language as parameters. The above example can be executed as follows:

```
$etur Example.dat Example.txt 2
```

If the third parameter (language) is omitted, the Spanish language will be taken by the program.

4.- Input data file

This file contains all the necessary data that define the problem, required for solving it.

Basically, it contains five data groups, all of them are necessary. They will be numbered from 1 to 5 and must be in order.

All the numerical data contained in this file will be read in a free format. The variables will be separated by one or more blank spaces.

The numbering of each group will be included in the first column. Listed below are the contents of each data group:

GROUP 1: Title of the problem:

Four lines containing the title of the problem to be solved.

Line 1 -----> 1 First line of the title

Line 2 -----> Second line of the title

Line 3 -----> Third line of the title

Line 4 -----> Fourth line of the title

GROUP 2: Fasteners data:

This group has N+1 lines with the following content:

Line 1 -----> 2 "Optional commentary"

Line 2 -----> N D(1) Er(1) tr₁(1) tr₂(1) Pitch(1)

Line 3 -----> D(2) Er(2) tr₁(2) tr₂(2) Pitch (1)

.....

.....

Line N+1 ---> D(N) Er(N) tr₁(N) tr₂(N) Pitch (N)

Where: N = Number of fasteners. This number will be also the numbers of elements to be considered for both panels.

D(i) = Is the diameter of the fastener "i".

Er(i) = Is the Young modulus of the fastener "i".

tr₁(i) = Thickness of the panel 1 at the location of fastener "i".

tr₂(i) = Thickness of the panel 2 at the location of fastener "i".

Pitch(i) = Is the distance between fasteners “i” and “i+1”. If the fastener “i” is the last, Pitch(i) is the distance from the fastener to the section considered as clamped.

GROUP 3: Panel 1 data:

This group has N+1 lines with the following content:

Line 1 -----> 3 “Optional commentary”

Line 2 -----> $E_1(1)$ $t_1(1)$ $W_1(1)$ $\alpha_1(1)$

Line 3 -----> $E_1(2)$ $t_1(2)$ $W_1(2)$ $\alpha_1(2)$

Line N+1 ---> $E_1(N)$ $t_1(N)$ $W_1(N)$ $\alpha_1(N)$

Where $E_1(i)$ = Is the Young modulus of the panel 1, for the segment “i”, located between fasteners “i” and “i+1”.

$t_1(i)$ = Is the thickness of the segment “i”, in the panel 1.

$W_1(i)$ = Is the equivalent width of the segment “i”, in the panel 1.

$\alpha_1(i)$ = Is the thermal expansion coefficient for the segment “i”, in the panel 1.

GROUP 4: Panel 2 data:

This group has also N+1 lines. Its content is the same as for group 3, but for the panel 2. It is:

Line 1 -----> 4 “Optional commentary”

Line 2 -----> $E_2(1)$ $t_2(1)$ $W_2(1)$ $\alpha_2(1)$

Line 3 -----> $E_2(2)$ $t_2(2)$ $W_2(2)$ $\alpha_2(2)$

Line N+1 ---> $E_2(N)$ $t_2(N)$ $W_2(N)$ $\alpha_2(N)$

Where $E_2(i)$ = Is the Young modulus of the panel 2, for the segment “i”, located between fasteners “i” and “i+1”.

$t_2(i)$ = Is the thickness of the segment “i”, in the panel 2.

$W_2(i)$ = Is the equivalent width of the segment “i”, in the panel 2.

$\alpha_2(i)$ = Is the thermal expansion coefficient for the segment “i”, in the panel 2.

GROUP 5: Increment of temperature:

This group has two lines with the following structure:

Line 1 -----> 5 “Optional commentary”

Line 2 -----> ΔT

Where: ΔT = Increment of temperature that the joint is subjected to. It is calculated as follows:

$$\Delta T = (\text{Working temperature}) - (\text{Assembly temperature})$$

5.- Output results

This section includes the description of the output result file as well as the possible error messages.

5.1 Output result file

This file provides the following information:

- **Summary of the input data:**
 - = Title.
 - = Fasteners data.
 - = Panel 1 data.
 - = Panel 2 data.
 - = Increment of temperature.
- **Results for fasteners:**
 - = Stiffness of each fastener.
 - = Load transferred by each fastener.
- **Results for panels:**
 - = Internal forces in each segment for panels 1 and 2.
 - = Displacements of nodes for panels 1 and 2.

5.2 Errors:

When the program detects an error in the input data, performing the analysis or writing the results, it writes the corresponding message in the console and aborts the execution of the program.

The error messages that might appear in the console are listed below:

Error messages. English output:

ERROR: The input file cannot be opened

ERROR: Erroneous group number in the input file

ERROR: Group 1 missing in input file (Title)

ERROR: End of file while reading group 1 in input file (Title)

ERROR: Group 2 missing in input file (Fasteners)

ERROR: End of File or error while reading group 2 in input file (Fasteners)

ERROR: The number of fasteners exceeds the maximum of the program

ERROR: The fastener diameter has to be greater than zero

ERROR: The Young modulus of the fastener has to be greater than zero

ERROR: The thickness of the panel 1 at the fastener has to be greater than zero

ERROR: The thickness of the panel 2 at the fastener has to be greater than zero

ERROR: The distance to the next fastener has to be greater than zero

ERROR: Group 3 missing in input file (Panel 1)

ERROR: End of file or error while reading group 3 in input file (Panel 1)

ERROR: The Young modulus of the panel 1 has to be greater than zero

ERROR: The thickness of the panel 1 has to be greater than zero

ERROR: The equivalent width of the panel 1 has to be greater than zero

ERROR: Group 4 missing in input file (Panel 2)

ERROR: End of file or error while reading group 4 in input file (Panel 2)

ERROR: The Young modulus of the panel 2 has to be greater than zero

ERROR: The thickness of the panel 2 has to be greater than zero

ERROR: The equivalent width of the panel 2 has to be greater than zero

ERROR: Group 5 missing in input file (Temperature)

ERROR: End of file or error while reading group 5 in input file (Temperature)

ERROR: The problem cannot be solved (singular matrix)

ERROR: The output file cannot be opened

ERROR: Writing error in the output file

Error messages. Spanish output:

ERROR: No se puede abrir el fichero de entrada

ERROR: Número de grupo erróneo en el fichero de entrada

ERROR: Grupo 1 ausente en el fichero de entrada (Título)

ERROR: Fin de fichero mientras se leía el grupo 1 en el fichero de entrada (Título)

ERROR: Grupo 2 ausente en el fichero de entrada (Remaches)

ERROR: Fin de fichero o error mientras se leía el grupo 2 (Remaches)

ERROR: El número de remaches excede el máximo del programa

ERROR: El diámetro del remache tiene que ser mayor que cero

ERROR: El módulo de elasticidad del remache tiene que ser mayor que cero

ERROR: El espesor del panel 1 en el remache tiene que ser mayor que cero

ERROR: El espesor del panel 2 en el remache tiene que ser mayor que cero

ERROR: La distancia al siguiente remache tiene que ser mayor que cero

ERROR: Grupo 3 ausente en el fichero de entrada (Panel 1)

ERROR: Fin de fichero o error mientras se leía el grupo 3 (Panel 1)

ERROR: El módulo de elasticidad del panel 1 tiene que ser mayor que cero

ERROR: El espesor del panel 1 tiene que ser mayor que cero

ERROR: La anchura equivalente del panel 1 tiene que ser mayor que cero

ERROR: Grupo 4 ausente en el fichero de entrada (Panel 2)

ERROR: Fin de fichero o error mientras se leía el grupo 4 (Panel 2)

ERROR: El módulo de elasticidad del panel 2 tiene que ser mayor que cero

ERROR: El espesor del panel 2 tiene que ser mayor que cero

ERROR: La anchura equivalente del panel 2 tiene que ser mayor que cero

ERROR: Grupo 5 ausente en el fichero de entrada (Temperatura)

ERROR: Fin de fichero o error mientras se leía el grupo 5 (Temperatura)

ERROR: No se puede resolver el problema (matriz singular)

ERROR: No se puede abrir el fichero de salida

ERROR: Error de escritura en el fichero de salida

6.- Windows application

An application has been developed in Visual Basic for Windows environment. The original routines of the program for input data, calculation and output results have been included in a Dynamic Link Library, which is used by this application. Therefore the same results will be obtained, with only differences due to rounding.

Since this is a Windows application, it must be installed in the user's PC before it can be executed. For installation, please consult with the system manager,

The application can be executed by "double click" in the file **Etur.exe**, located in the PC's server, or by means of a **link** in the "Desktop" or in the "Start Menu".

The input data can be retrieved from an input data file or directly typed in the application form. The form has the following appearance:

Title:

Program ETUR. Example number.

Five fasteners of titanium.

Panel 1 of steel.

Panel 2 of aluminium.

Joint data:

No. of fasteners 5

	Fasteners					Panel 1				Panel 2			
	Diam	Er	Thick 1	Thick 2	Pitch	E	Thicknes	Width	Coef	E	Thicknes	Width	Coef
1	5.0	110000.	1.0	4.0	20.0	196000.	1.0	25.0	1.30E-0	73000.0	4.0	80.0	2.30E-0
2	5.0	110000.	1.0	4.0	20.0	196000.	1.0	25.0	1.30E-0	73000.0	4.0	80.0	2.30E-0
3	5.0	110000.	1.0	4.0	20.0	196000.	1.0	25.0	1.30E-0	73000.0	4.0	80.0	2.30E-0
4	5.0	110000.	1.0	4.0	20.0	196000.	1.0	25.0	1.30E-0	73000.0	4.0	80.0	2.30E-0
5	5.0	110000.	1.0	4.0	10.0	196000.	1.0	25.0	1.30E-0	73000.0	4.0	80.0	2.30E-0

Increment of temperature: Working Temperature - Assembly Temperature 80.0

Input file:

Language:

☐ Español ☒ English

Program execution

Figure 2: Input data form.

The program can be executed clicking the button “Program execution”. Then the application solves the problem and shows the results in the following form:

Etur: Results

Loads in fasteners and elements:

	Fasteners		Panel 1		Panel 2	
	Stiffness	Load	Force	Displacement	Force	Displacement
1	3.311E+04	1024.5	1024.5	1.275E-01	0.0	1.585E-01
2	3.311E+04	662.2	1686.8	1.026E-01	0.0	1.226E-01
3	3.311E+04	408.2	2095.0	7.487E-02	0.0	8.720E-02
4	3.311E+04	221.0	2316.0	4.552E-02	0.0	5.220E-02
5	3.311E+04	69.9	2385.8	1.527E-02	0.0	1.738E-02

Return to the data Write the results Exit program

Figure 3: Output results form.

It is now possible to write the output results in a file, by clicking on the corresponding button (the mid one of the three buttons located at the bottom of the form).

7.- Examples

Input data and obtained results are presented for the four following examples.

7.1 Example 1

The case to solve is a stiffened panel. The joint to be analyzed is the attachment of a stringer to the panel. The geometry of the panel is shown on next figure:

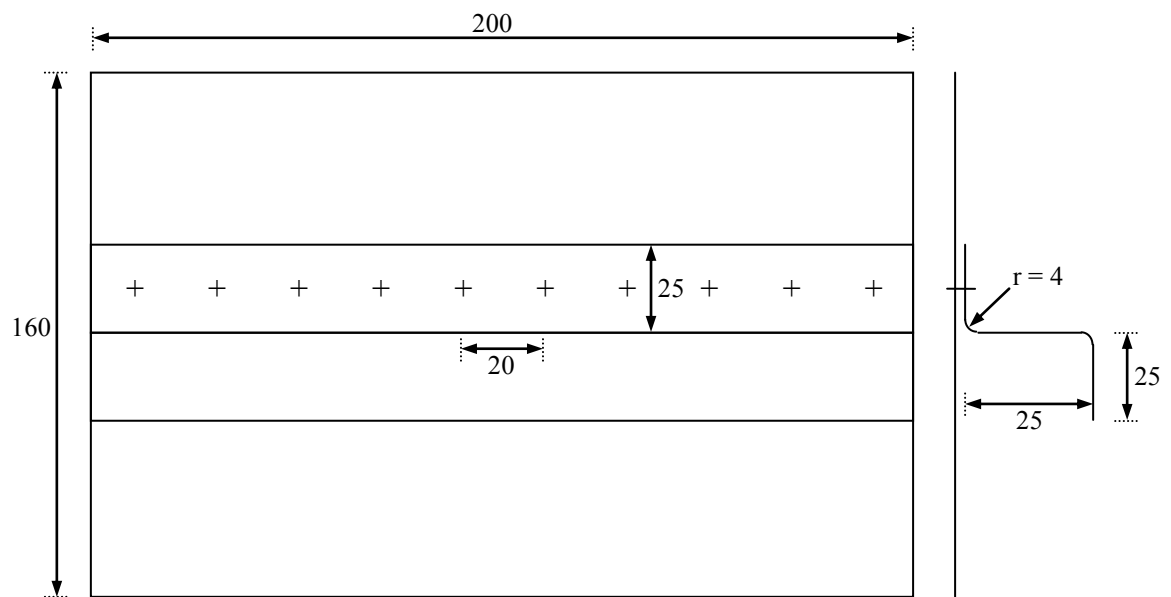


Figure 4: Example 1. Geometry.

The panel is an aluminium sheet meanwhile the stringer is a steel folded sheet. The panel has a thickness of 4 mm, the stringer thickness is 1 mm and all fasteners, made of titanium, have a diameter of 5 mm.

Due to the symmetry of the panel, it can be considered a half part, split in five segment, with an end clamped and the other free, as shown below:

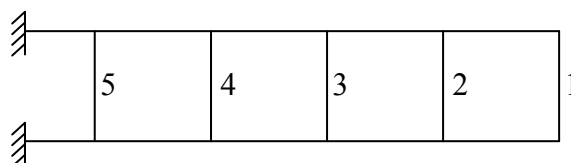


Figure 5: Example 1. Idealization.

The stringer will be idealized as panel 1 and the skin, panel 2. The equivalent width of each segment will be considered the total width around the mid line for the stringer (71.6 mm) and the total width for the panel (160 mm).

The corresponding input data and output results files are shown on next pages:

Input file:

```

1 Program ETUR. Example number 1.
  Five fasteners of titanium.
  Panel 1 of steel.
  Panel 2 of aluminium.
2 Fasteners: N  D  Er  Tr1  Tr2  Pitch
  5  5.0  110000.0  1.0  4.0  20.0
      5.0  110000.0  1.0  4.0  20.0
      5.0  110000.0  1.0  4.0  20.0
      5.0  110000.0  1.0  4.0  20.0
      5.0  110000.0  1.0  4.0  10.0
3 Panel 1: E1  T1  W1  A1
  196000.0  1.0  71.6  1.3E-05
  196000.0  1.0  71.6  1.3E-05
  196000.0  1.0  71.6  1.3E-05
  196000.0  1.0  71.6  1.3E-05
  196000.0  1.0  71.6  1.3E-05
4 Panel 2: E2  T2  W2  A2
  73000.0  4.0  160.0  2.3E-05
  73000.0  4.0  160.0  2.3E-05
  73000.0  4.0  160.0  2.3E-05
  73000.0  4.0  160.0  2.3E-05
  73000.0  4.0  160.0  2.3E-05
5 Delta of temperature
  80.0

```

Output file:

```
*****
*
*          <<< ETUR >>>
*
*****
* Calculation of thermal loads
* in a fastened joint
*****
```

Title:

=====

Program ETUR. Example number 1.
Five fasteners of titanium.
Panel 1 of steel.
Panel 2 of aluminium.

Fasteners data:

=====

Num	Diameter	Modulus	Thickn. 1	Thickn. 2	Pitch
---	-----	-----	-----	-----	-----
1	5.0	110000.0	1.0	4.0	20.0
2	5.0	110000.0	1.0	4.0	20.0
3	5.0	110000.0	1.0	4.0	20.0
4	5.0	110000.0	1.0	4.0	20.0
5	5.0	110000.0	1.0	4.0	10.0

Panels data:

=====

Panel 1:

-----:

Num	Modulus	Thickness	Width	Coef.
---	-----	-----	-----	-----
1	196000.0	1.0	71.6	1.300E-05
2	196000.0	1.0	71.6	1.300E-05
3	196000.0	1.0	71.6	1.300E-05
4	196000.0	1.0	71.6	1.300E-05
5	196000.0	1.0	71.6	1.300E-05

Panel 2:

-----:

Num	Modulus	Thickness	Width	Coef.
---	-----	-----	-----	-----
1	73000.0	4.0	160.0	2.300E-05
2	73000.0	4.0	160.0	2.300E-05
3	73000.0	4.0	160.0	2.300E-05
4	73000.0	4.0	160.0	2.300E-05
5	73000.0	4.0	160.0	2.300E-05

Increment of temperature:

=====

Delta = 80.00

Output file: (cont.)

Loads in fasteners and elements:

=====

Num	Fastener		Panel 1		Panel 2	
	Stiffness	Load	Force	Displ.	Force	Displ.
1	3.311E+04	1554.3	1554.3	1.129E-01	-1554.3	1.598E-01
2	3.311E+04	1119.9	2674.1	8.986E-02	-2674.1	1.237E-01
3	3.311E+04	754.2	3428.3	6.525E-02	-3428.3	8.802E-02
4	3.311E+04	434.7	3863.0	3.956E-02	-3863.0	5.269E-02
5	3.311E+04	142.0	4005.0	1.325E-02	-4005.0	1.754E-02

7.2 Example 2

The example includes an analysis of the joint of a machined angle, made in titanium, attached to a panel, which is in aluminium. The angle has a constant thickness of 5 mm meanwhile the panel is an 8 mm thickness sheet.

Next figure shows a sketch of the joint to be analysed:

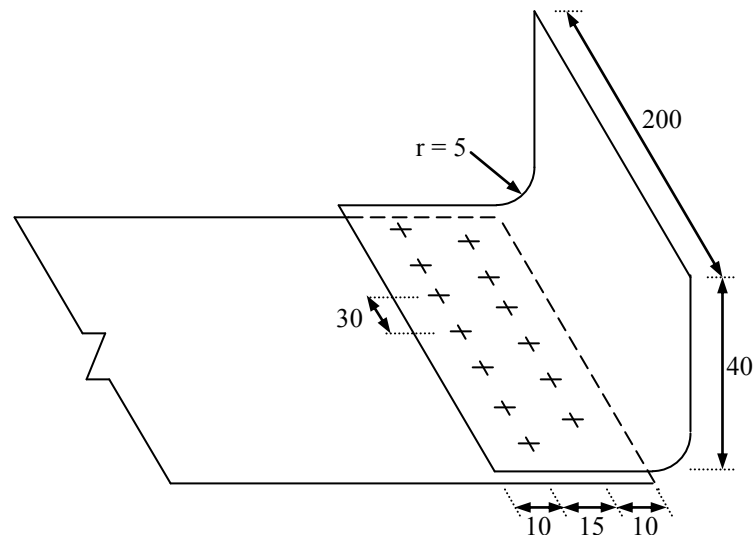


Figure 6: Example 2. Geometry.

Due to the symmetrical configuration of this case, the fastener in the plane of symmetry does not work, which implies idealized joint is equivalent to a line with 6 fasteners with a 15 mm pitch.

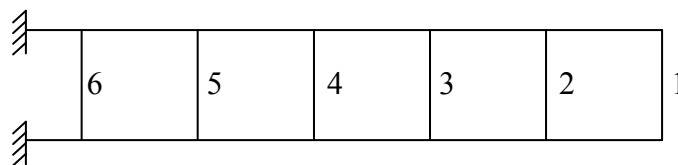


Figure 7: Example 2. Idealization.

The angle will be idealized as panel 1 and the skin, panel 2. The equivalent width of each segment will be considered the total width around the mid line for the angle (77.9 mm) and the same width for the panel. All fasteners have the same diameter: 5 mm.

The corresponding input data and output results files are shown on next pages:

Input file:

[illegible]

Output file: (cont.)

```
*****
*
*          <<< ETUR >>>
*
*****
* Calculation of thermal loads
* in a fastened joint
*****
```

Title:

=====

Program ETUR. Example number 2.

Six fasteners of titanium.

Panel 1 of titanium.

Panel 2 of aluminium.

Fasteners data:

=====

Num	Diameter	Modulus	Thickn. 1	Thickn. 2	Pitch
---	-----	-----	-----	-----	-----
1	5.00	110000.0	5.00	8.00	15.00
2	5.00	110000.0	5.00	8.00	15.00
3	5.00	110000.0	5.00	8.00	15.00
4	5.00	110000.0	5.00	8.00	15.00
5	5.00	110000.0	5.00	8.00	15.00
6	5.00	110000.0	5.00	8.00	15.00

Panels data:

=====

Panel 1:

-----:

Num	Modulus	Thickness	Width	Coef.
---	-----	-----	-----	-----
1	110000.0	5.00	77.90	9.000E-06
2	110000.0	5.00	77.90	9.000E-06
3	110000.0	5.00	77.90	9.000E-06
4	110000.0	5.00	77.90	9.000E-06
5	110000.0	5.00	77.90	9.000E-06
6	110000.0	5.00	77.90	9.000E-06

Panel 2:

-----:

Num	Modulus	Thickness	Width	Coef.
---	-----	-----	-----	-----
1	73000.0	8.00	77.90	2.300E-05
2	73000.0	8.00	77.90	2.300E-05
3	73000.0	8.00	77.90	2.300E-05
4	73000.0	8.00	77.90	2.300E-05
5	73000.0	8.00	77.90	2.300E-05
6	73000.0	8.00	77.90	2.300E-05

Output file: (cont.)

Increment of temperature:

=====

Delta = 80.00

Loads in fasteners and elements:

=====

Num	Fastener		Panel 1		Panel 2	
	Stiffness	Load	Force	Displ.	Force	Displ.
1	4.369E+04	3083.4	3083.4	8.037E-02	-3083.4	1.509E-01
2	4.369E+04	2441.0	5524.3	6.849E-02	-5524.3	1.244E-01
3	4.369E+04	1871.1	7395.4	5.575E-02	-7395.4	9.858E-02
4	4.369E+04	1356.7	8752.2	4.236E-02	-8752.2	7.342E-02
5	4.369E+04	882.7	9634.9	2.850E-02	-9634.9	4.870E-02
6	4.369E+04	434.9	10069.8	1.433E-02	-10069.8	2.428E-02

7.3 Example 3

The case to solve is a fitting attached to the skin. It is a representative case of a double shear joint. The geometry of the junction is shown below:

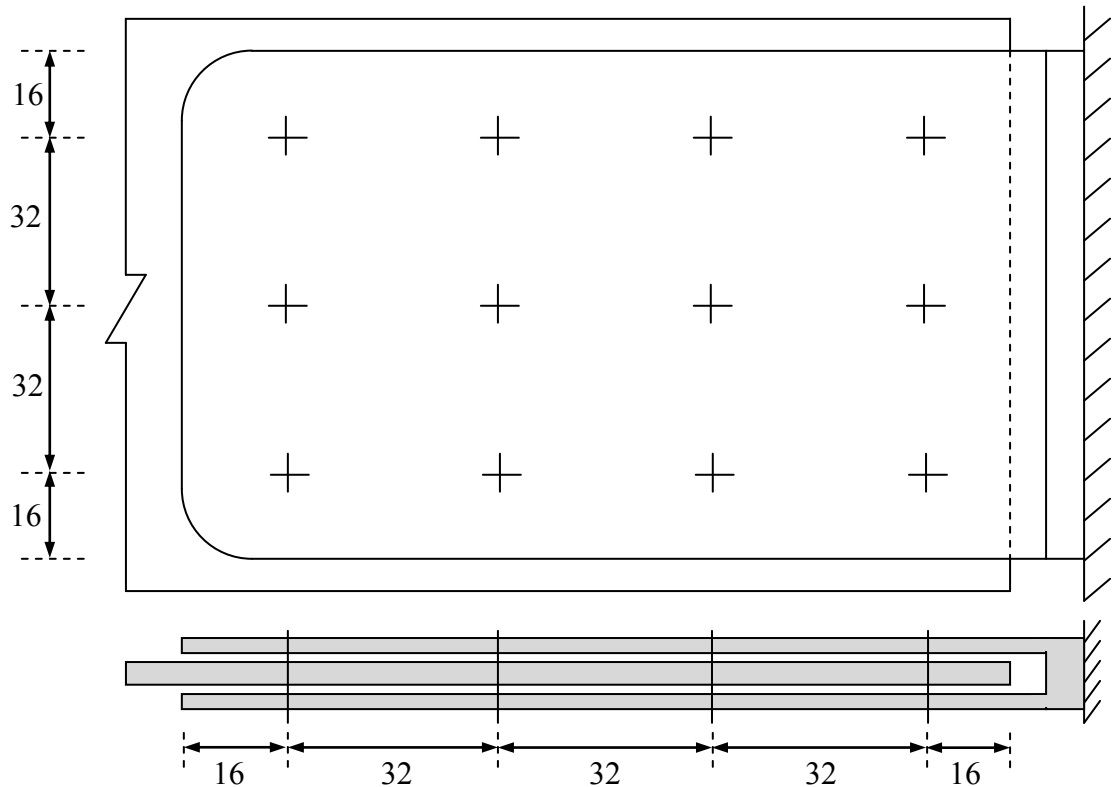


Figure 8: Example 3. Geometry.

The panel is made in CFC and the fitting is a titanium alloy. The skin thickness is 16 mm and each one of the fitting plates has 6 mm of thickness. The fasteners have a diameter of 8 mm and all they are made of titanium.

To idealize the joint, it is necessary to take into account that it is a double shear joint with three row of fasteners. Both aspects can be solved as follows:

- Due to the symmetry, the joint can be split in two single shear joints, taking one plate of the fitting (6 mm) and the half part of the skin thickness (8 mm).
- It can be assumed that the behaviour of each row of four fasteners is the same. Therefore, only one row will be considered. The equivalent width of each element will be 32 mm.

Taking into account these hypotheses, the joint can be idealized as shown on next figure:

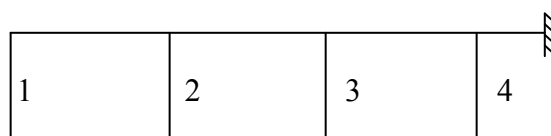


Figure 9: Example 3. Idealization.

The corresponding input data and output results files are shown on next pages:

Input file:

```

1 Program ETUR. Example number 3.
  Four fasteners of titanium.
  Panel 1 of titanium.
  Panel 2 of CFC.
2 Fasteners: N   D   Er   Tr1   Tr2   Pitch
  4  8.0  110000.0  6.0   8.0   32.0
      8.0  110000.0  6.0   8.0   32.0
      8.0  110000.0  6.0   8.0   32.0
      8.0  110000.0  6.0   8.0   20.0
3 Panel 1: E1   T1   W1   A1
  110000.0  6.0  32.0  9.00E-06
  110000.0  6.0  32.0  9.00E-06
  110000.0  6.0  32.0  9.00E-06
  110000.0  6.0  32.0  9.00E-06
4 Panel 2: E2   T2   W2   A2
  62190.0  8.0  32.0  2.33E-06
  62190.0  8.0  32.0  2.33E-06
  62190.0  8.0  32.0  2.33E-06
  62190.0  0.0  32.0  2.33E-06
5 Delta of temperature
  80.0

```

Output file:

```
*****
*
*          <<< ETUR >>>
*
*****
* Calculation of thermal loads
* in a fastened joint
*****
```

Title:

=====

Program ETUR. Example number 3.
Four fasteners of titanium.
Panel 1 of titanium.
Panel 2 of CFC.

Fasteners data:

=====

Num	Diameter	Modulus	Thickn. 1	Thickn. 2	Pitch
---	-----	-----	-----	-----	-----
1	8.00	110000.0	6.00	8.00	32.00
2	8.00	110000.0	6.00	8.00	32.00
3	8.00	110000.0	6.00	8.00	32.00
4	8.00	110000.0	6.00	8.00	20.00

Panels data:

=====

Panel 1:

-----:

Num	Modulus	Thickness	Width	Coef.
---	-----	-----	-----	-----
1	110000.0	6.00	32.00	9.000E-06
2	110000.0	6.00	32.00	9.000E-06
3	110000.0	6.00	32.00	9.000E-06
4	110000.0	6.00	32.00	9.000E-06

Panel 2:

-----:

Num	Modulus	Thickness	Width	Coef.
---	-----	-----	-----	-----
1	62190.0	8.00	32.00	2.330E-06
2	62190.0	8.00	32.00	2.330E-06
3	62190.0	8.00	32.00	2.330E-06
4	62190.0	0.00	32.00	2.330E-06

Increment of temperature:

=====

Delta = 80.00

Output file: (cont.)

Loads in fasteners and elements:

=====

Num	Fastener		Panel 1		Panel 2	
	Stiffness	Load	Force	Displ.	Force	Displ.
1	6.693E+04	-1232.9	-1232.9	7.734E-02	1232.9	5.892E-02
2	6.693E+04	-381.0	-1614.0	5.617E-02	1614.0	5.047E-02
3	6.693E+04	381.0	-1232.9	3.557E-02	1232.9	4.126E-02
4	6.693E+04	1232.9	0.0	1.440E-02	0.0	3.282E-02

7.4 Example 4

This case is a multiple joint representing a spar web interrupted in an intersection by a rib. The joint between both sides of the spar web and the rib end is by means of two auxiliary plates, as shown on next figure:

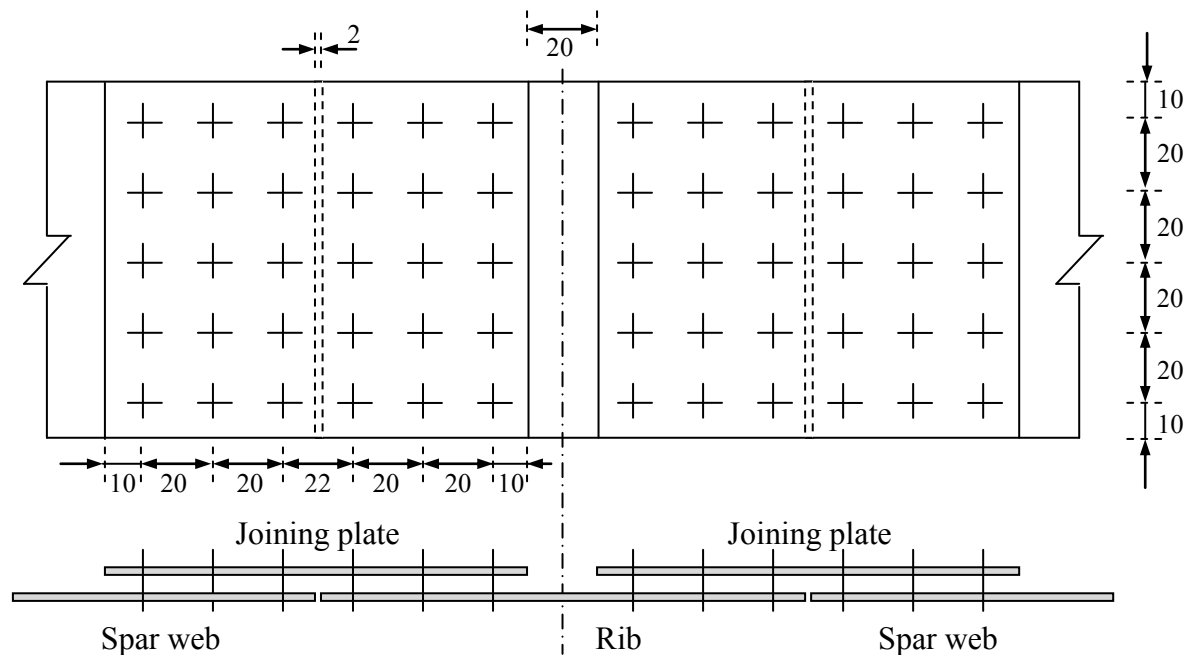


Figure 10: Example 4. Geometry.

The spar web is CFC, thickness 6 mm, the rib is aluminium, thickness 6 mm, the joining plates are titanium, thickness 4 mm, and the bolts are inconel with a diameter of 5 mm.

Due to the symmetry of the joint, it can be considered the half part, split in six segments, with an end clamped and the other free, as shown below:

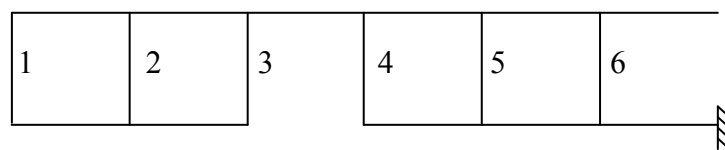


Figure 11: Example 4. Idealization.

Due to discontinuity between fasteners 3 and 4, the segment 3 of the lower panel does not carry out load. Therefore it will be considered a zero thickness segment. For the same reason, the upper panel after fastener 6 will have zero thickness.

The corresponding input data and output results files are shown on next pages:

Input file:

```

1 Program ETUR. Example number 4.
  Six fasteners of inconel.
  Panel 1 of CFC/aluminium.
  Panel 2 of titanium.
2 Fasteners: N   D   Er   Tr1   Tr2   Pitch
  6  5.0  196000.0  6.0   4.0   20.0
    5.0  196000.0  6.0   4.0   20.0
    5.0  196000.0  6.0   4.0   22.0
    5.0  196000.0  6.0   4.0   20.0
    5.0  196000.0  6.0   4.0   20.0
    5.0  196000.0  6.0   4.0   20.0
3 Panel 1: E1   T1   W1   A1
  110000.0  4.0  20.0  9.00E-06
  110000.0  4.0  20.0  9.00E-06
  110000.0  4.0  20.0  9.00E-06
  110000.0  4.0  20.0  9.00E-06
  110000.0  4.0  20.0  9.00E-06
  110000.0  0.0  20.0  9.00E-06
4 Panel 2: E2   T2   W2   A2
  62190.0  6.0  20.0  2.33E-06
  62190.0  6.0  20.0  2.33E-06
  62190.0  0.0  22.0  2.33E-06
  73000.0  6.0  20.0  2.30E-05
  73000.0  6.0  20.0  2.30E-05
  73000.0  6.0  20.0  2.30E-05
5 Delta of temperature
  80.0

```

Output file:

```
*****
*
*          <<< ETUR >>>
*
*****
* Calculation of thermal loads
* in a fastened joint
*****
```

Title:

=====

Program ETUR. Example number 4.
Six fasteners of inconel.
Panel 1 of CFC/aluminium.
Panel 2 of titanium.

Fasteners data:

=====

Num	Diameter	Modulus	Thickn. 1	Thickn. 2	Pitch
---	-----	-----	-----	-----	-----
1	5.00	196000.0	4.00	6.00	20.00
2	5.00	196000.0	4.00	6.00	20.00
3	5.00	196000.0	4.00	6.00	22.00
4	5.00	196000.0	4.00	6.00	20.00
5	5.00	196000.0	4.00	6.00	20.00
6	5.00	196000.0	4.00	6.00	20.00

Panels data:

=====

Panel 1:

-----:

Num	Modulus	Thickness	Width	Coef.
---	-----	-----	-----	-----
1	110000.0	4.00	20.00	9.000E-06
2	110000.0	4.00	20.00	9.000E-06
3	110000.0	4.00	20.00	9.000E-06
4	110000.0	4.00	20.00	9.000E-06
5	110000.0	4.00	20.00	9.000E-06
6	110000.0	0.00	20.00	9.000E-06

Panel 2:

-----:

Num	Modulus	Thickness	Width	Coef.
---	-----	-----	-----	-----
1	62190.0	6.00	20.00	2.330E-06
2	62190.0	6.00	20.00	2.330E-06
3	62190.0	0.00	20.00	2.330E-06
4	73000.0	6.00	20.00	2.300E-05
5	73000.0	6.00	20.00	2.300E-05
6	73000.0	6.00	20.00	2.300E-05

Output file: (cont.)

Increment of temperature:

=====

Delta = 80.00

Loads in fasteners and elements:

=====

Num	Fastener		Panel 1		Panel 2	
	Stiffness	Load	Force	Displ.	Force	Displ.
1	4.935E+04	-423.2	-423.2	1.307E-01	423.2	1.221E-01
2	4.935E+04	0.0	-423.2	1.173E-01	423.2	1.173E-01
3	4.935E+04	423.2	0.0	1.038E-01	0.0	1.124E-01
4	5.281E+04	953.5	953.5	8.799E-02	-953.5	1.060E-01
5	5.281E+04	0.0	953.5	7.142E-02	-953.5	7.142E-02
6	5.281E+04	-953.5	0.0	5.486E-02	0.0	3.680E-02